Boon Academy — Intervention Triage

A facilitator-mediated, closed-loop system for getting help to failing students before the next quiz — technical report

AI Systems Engineer case study

June 2026

Boon Academy already knows *who* is failing; only ~30% of those students get any intervention before the next quiz. The bottleneck is not detection — it is **triage, action, and never knowing whether the help worked**. This system produces a **two-tier action queue** per facilitator — calls capped at what a human can do, drafted WhatsApp messages uncapped, so every quiz-failer leaves with a ready action — by fusing a transparent rule engine (the numbers, backtested at **AUC 0.91** against Quiz-1 outcomes) with an LLM that *reads* the Arabic facilitator notes (measured at $\kappa = 0.80$ against 75 human gold labels). The measurement layer claims only what the data supports: only **44%** of quiz-failers have any logged contact since the quiz, and of the contacts that exist, just 11% were followed by re-engagement — a descriptive gap (no control group) that motivates the one-field outcome log in the roadmap. Every design choice is grounded in the early-warning and behavioural-nudge evidence base and stated with its limitations.

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1 The problem and the diagnosis

The brief frames a fictional "Boon Academy": students attend in person, on-site **facilitators** run each classroom, and teachers instruct remotely. When a student falls behind the facilitator is supposed to intervene (call the parent, 1-on-1 tutoring, a motivational message), but only ~30% of failing students are reached before the next quiz, and the org is scaling 1,000 → 5,000 students.

The real problem is not an AI problem

A facilitator running a full classroom cannot act on a wall of red flags — it is **cognitive overload and decision fatigue**. If she can make three calls today, the system must tell her *which three*, *why*, and *what to say*. Three deeper truths shape the design: (1) the signal is split between LMS *numbers* and the facilitators' own free-text *notes*, and nobody synthesises them; (2) attendance is a deceptive proxy; (3) a logged note $\neq$ an effective intervention.

2 The data

Three CSVs joined on `student_id` only (names written inside notes do not match the roster). 200 students across 5 campuses and 8 facilitators; 10 active class-days (1–14 Oct 2025, weekends absent); Quiz 1 on day 10; the system's "today" is a parameter (`AS_OF_DATE`=2025-10-14), never `now()`, so runs are reproducible.

metadata `student_id, student_name, campus_id, facilitator_email, grade, parent_phone, target_score, learning_track`

daily_metrics `student_id, date, session_attended_min, practice_questions, last_quiz_score, days_until_next_quiz`

notes `note_id, student_id, facilitator_email, date, note_text` (180 notes, 100% Arabic free text)

2.1 The data traps (handled once, in ingest)

Real operational data is messy; each trap is neutralised in one place so the rest of the pipeline can trust the record.

- **Name mismatch** — join on `student_id` only.
- **Carried-forward quiz** — `last_quiz_score` repeats across days; it is one event (taken at the quiz date), not a daily score.
- **Quiz 0 that is an absence** — a `0` with a *blank* session on quiz day means "absent during the quiz", not a real failing score (student S145).
- **Corrupt `target_score`** — a whole Remedial cohort has targets below the pass mark (e.g. 12); never used as a signal.
- **Dirty phones** — a missing `+`, an email in the phone field — normalised and flagged, never echoed raw (privacy).
- **Blank cells** — kept distinct from a real `0.0`.

2.2 Data insights

Table 1: Quiz-1 failure by learning track — track is nearly destiny. (One further Remedial "0" is an exam-day *absence*, not a failure — counted separately.)

Track	Failed / total	Fail rate
Accelerated	0 / 14	**0%**
Standard	15 / 128	12%
Remedial	51 / 58	**88%**
All	66 / 200	33%

- **Coverage & blind spots:** only **75/200 (38%)** students have any note; **125 (62%) were never contacted**, including **20 who failed the quiz and have no note at all** — invisible to any notes-based view.
- **Attendance is deceptive:** ~15 students attend ~88 min every day, do almost no practice, and score in the 50s–low-60s. An attendance-only triage rates them flawless; only practice + quiz catch them. Hence practice is a *separate* axis from attendance.
- **More practice can mean more risk:** one student did 120 practice questions in a single day (cramming) and still failed — so raw volume is not rewarded; the engine uses the spike-robust *median* and flags single-day spikes.
- **Notes are reactive:** 74 of 180 notes (41%) land on the single final day, with secondary spikes on days 3–4 (enrollment) and day 13. Notes are written *after* the fact, not as early warning.

2.3 How facilitators differ in note-taking

Note-taking is far from uniform, and the differences map onto the overload story directly.

Table 2: Per-facilitator note-taking. Notes are short (median 12 words, range 2–23) and uniformly Arabic; coverage and verbosity vary widely.

Facilitator	Roster	Noted	Coverage	Notes	Median words
facilitator1	20	9	45%	27	13
facilitator2	20	8	40%	19	**14**
facilitator3	35	12	34%	26	14
facilitator4	22	11	**50%**	24	10
facilitator5	20	9	45%	23	12
facilitator6	24	9	38%	19	10
facilitator7	21	9	43%	21	11
facilitator8	38	8	**21%**	21	**9**

The overload is visible in the data

The two facilitators with the *largest* rosters (38 and 35 students) have the two *lowest* note-coverage rates (21% and 34%) and, for the 38-student case, the *tersest* notes (median 9 words). Bigger caseload → less documentation → more students falling through. This is precisely the capacity problem the system is built to relieve — and the reason the queue caps **only the expensive actions** per facilitator (drafted messages scale with need, facilitator minutes do not), rather than dumping a global list.

Practical consequences for the LLM: notes are *short and dialectal* (12 words median), so the reader must extract a lot from little and tolerate Najdi/Hijazi spelling, code-switching, and abbreviations; and because *count/recency* of notes varies by facilitator habit rather than by student need, note metadata is a poor risk signal — another reason to read *content*, not count.

2.4 The demo cast — where numbers and notes disagree

The dataset is built around students whose numbers and notes point in opposite directions; these drive the demo and the acceptance tests.

Table 3: Representative students. The point of reading notes is the right-hand column.

ID	Numbers say	Notes say	Correct call
S023	passing (88)	vanished, no reply	**Critical** (silent dropout)
S005	passing (85)	"worried, needs immediate follow-up"	**Critical** (failing intervention)
S145	quiz 0	"big transformation, dad engaged"	**Low** (the 0 is an absence)
S017	passing (83)	"crying — family problems"	**Surface** (note-only crisis)
S199	quiz 9, target 12	(no note)	**Critical** (invisible at-risk)
S051	quiz 52	"practising regularly" (120-q cram)	**High** (do not demote)
S076	passing (84)	"excellent student"	**Leave alone**

3 Architecture

A deliberately small linear pipeline (~1,450 lines of Python; pandas + the Google `genai` SDK) with **one** LLM seam. Four objects flow through it:

$$\texttt{StudentRecord} \rightarrow \texttt{Risk} + \texttt{NoteState} \rightarrow \texttt{ActionRow} \rightarrow \texttt{queue}.$$

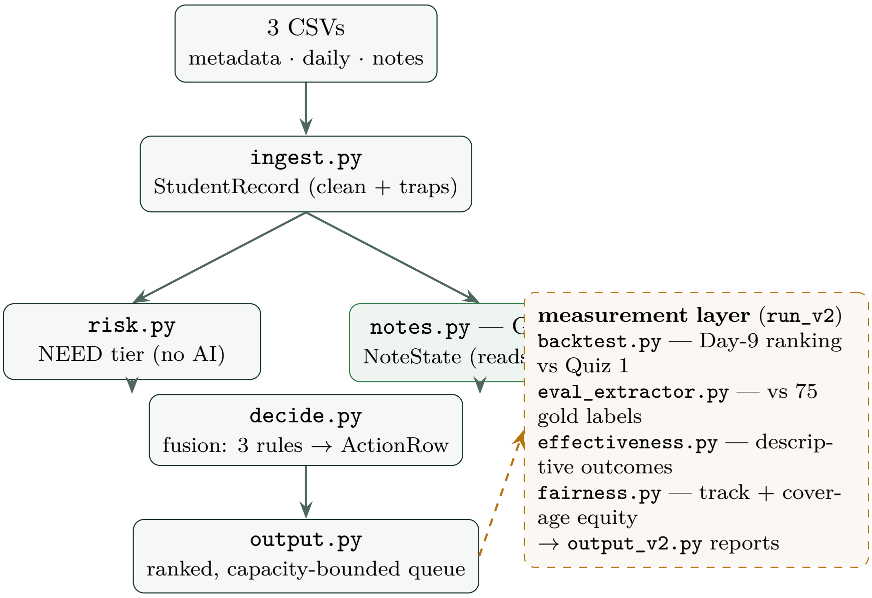

Figure 1: Data flow. `config.py` feeds constants everywhere; `pipeline.py` wires it; `main.py` is the CLI. The only LLM call is the green box.

Table 4: Files and responsibilities (~2,300 lines total).

File	Job	Lines
`config.py`	tunables, `.env` loader, the case calendar	73
`ingest.py`	3 CSVs → StudentRecord; every trap, once; dated notes	219
`risk.py`	NEED: 4 signals + cliff → tier + reasons (no AI)	84
`notes.py`	the LLM seam: Gemini reads threads → NoteState	281
`drafts.py`	template MSA drafts for the un-noted majority (no AI)	67
`decide.py`	fusion: NEED × STATE, 3 rule families → ActionRow	171
`output.py`	two-tier queue, KPI counters, contact log, CSV/JSON/HTML	368
`pipeline.py`	orchestrator: run / run_v2 / lift	114
`backtest.py`	Day-9 ranking backtest vs Quiz-1 outcomes	88
`eval_extractor.py`	note-reader vs the human gold labels	104
`effectiveness.py`	descriptive post-contact outcomes (+caveats)	135
`fairness.py`	track audit + facilitator coverage equity	86
`output_v2.py`	write the measurement reports	133
`tests/test_cast.py`	23 acceptance tests (traps, guards, KPI invariants)	193

4 The risk engine (NEED) — where a simple rule does the job

Deterministic and explainable on purpose: 14 days of data, no labelled outcomes, and a decision about children all argue for transparent rules over a black box. Four contributors, severity-ordered, each attaching a human-readable reason:

Coursework failed Quiz 1 (absence-flagged scores excluded).
Attendance chronically low recent attendance (only when it is *not* a cliff, to avoid double-counting).
Behaviour little *sustained* practice (median, so a cram spike is not credited).
Trajectory a strong attendant who *collapses* to near-zero in the last days — a silent dropout the quiz can never catch (S023).

Points map to a tier (Critical/High/Medium/Low). Notably, `target_score` is *excluded* (corrupt for a whole cohort) and `learning_track` is *not* an input (fairness, § 7.1).

5 The note-reader — the only place AI is used, and why

The one job a rule cannot do is read short, informal, dialectal Arabic prose. So the LLM is pointed at exactly one thing — the notes — for two tasks a rule can't: **reading** them into a structured state that changes the priority, and **drafting** the parent message. It never scores risk.

> **Extractor, not oracle**
>
> The LLM emits a small structured *state*; deterministic rules decide. This bounds the blast radius of any wrong read: it can escalate, demote, or flag-for-review, but it can never silently drop a failing student (the numbers floor always surfaces them), and a working-note can never demote a quiz-failing student (a guarded rule). The model proposes; the rules dispose.

5.1 Model choice

Gemini 2.5 Flash Lite, temperature 0, via the API behind a swappable `LLMBackend` interface (a `NoLLM` backend gives the rules-only baseline).

> **Why this model, and the PDPL path**
>
> Native structured output (`responseSchema`) guarantees schema-valid JSON — which kills the single biggest execution risk (a model emitting broken JSON from dialectal Arabic). It is cheap (~$0.0003/student) and fast. Because it is one isolated module, production with real minors' PII under PDPL swaps it for a KSA-region or on-prem Arabic model *without touching the pipeline.* A second finding shaped the message design: current models *read* Gulf dialect well but *generate* it poorly — so messages are drafted in warm, simple Modern Standard Arabic (the natural polite register for a Saudi family

message anyway), not synthetic colloquial.

5.2 The prompt

The system prompt (lightly abridged) — the schema *is* the architecture:

```
You read one facilitator's note THREAD about ONE student (short, informal Saudi-dialect
Arabic, chronological, each line dated [Day N]). You never see the metrics -- a rules
engine scores those. Report what the PROSE says, weighting the most recent days. JSON:
- state: improving | explained | needs_help | failing | refused | none
    improving was struggling, now clearly better / intervention working
    explained known, managed, temporary cause (illness, family event) AND the
              parent is aware -- monitoring is enough for now
    needs_help real problem, no working intervention yet (incl. "we must intervene")
    failing help WAS attempted and is NOT working -- incl. a vanished student
              with unreachable parents (the contact channel itself is failing)
    refused parent/student EXPLICITLY declined or pushed back on offered help
    none administrative or positive-only
- blocker: academic | motivation | family | health | logistics | unknown
- summary one plain sentence: what is going on
- root_cause e.g. gaming, family issue, unaware of program (or "")
- suggested_action call_parent | one_on_one | message | none
- draft_message a short, warm WhatsApp message in SIMPLE Modern Standard Arabic, to the
     PARENT,
    sent BY the named facilitator: cite ONE specific fact; be improvement-focused (not
        praise/blame);
    propose exactly ONE doable step (NEVER a meeting); 2-3 sentences.
- evidence an EXACT Arabic span from the notes -- REQUIRED for any non-"none" state
- concern low | neutral | worried | urgent
- confidence low | medium | high (use low when vague -- do not guess)
Be faithful. Do not invent facts.
```

Why every field earns its place

- **state** drives the fusion override — it is the thing a rule cannot derive.
- **evidence** (a verbatim Arabic span) enables a *faithfulness check*: the quote must be a substring of a real note, or it is dropped and confidence lowered — the anti-hallucination floor.
- **confidence** powers *abstention*: low-confidence reads go to a human-review lane, never auto-acted.
- **draft_message** encodes the behavioural-nudge evidence (§ 8): *parent-directed*, *improvement-focused*, *one doable step*, *no meeting* (the Saudi working-parent barrier), sent by the named facilitator (the “trusted local sender”).

Two few-shot examples teach the dialect mapping. A failing case (تلصتا نيترم لع مالا، ام تدر. هيلع هناقلق) maps to `state=failing, concern=urgent`; a turnaround (ريبك لوحت) maps to `state=on_track`. The drafted message stays warm MSA, e.g. مكيلع مالسلا، دمحأ بلاط رداق نكل ابتخالا لبق انًسّحت عقّوتن مونلل تباث تقو لع هعم قفتن ول ؛عوبسألا اذه عجارت هروضح.

6 The decision levels (fusion)

Three gating rules combine NEED (numbers) and STATE (notes); deliberately three, not a matrix — the whole point of reading notes is to correct the two cases where the numbers lie.

1. **Escalate** — `state` is *failing/refused*: push up; a recent contact must NOT lower risk if the intervention is failing (S005).
2. **Demote** — `state` is *on_track* **and** the numbers agree (not failing, not a cliff): lower it to free capacity (S145, S070). The "numbers agree" guard is what stops a soft note from rescuing a quiz-failing crammer (S051).
3. **Human review** — a low-confidence/vague note: surface for a human, never auto-act (S013).

Surfacing rule: a student is queued if the numbers flag them *or* the note is non-positive — the single `OR` that catches S017 (clean numbers, a hidden family crisis). The queue is then ranked into **two tiers**: calls/1-on-1s capped at the top-N per facilitator (facilitator time is the scarce resource), and ready-to-send drafts *uncapped* — the largest classroom has 16 quiz-failers, so any flat top-8 list would cap its coverage at 50% forever. Every surfaced student leaves with a message (the LLM's if there are notes, a metric-filled MSA template otherwise).

7 v2: closing the loop — measuring whether help worked

A ranked queue is necessary but, on its own, insufficient: the early-warning literature shows that identifying at-risk students reliably fails to improve outcomes unless the data drives a real, completed intervention (§ 8). v2 adds the measurement layer.

The headline: attempted $\neq$ effective — and today we cannot even tell

Every facilitator note is a dated contact; the metrics are daily. Descriptively, of **75 logged contact threads: only 11% were followed by re-engagement, 11% kept declining, and 13% were logged too late to act** (final-day notes). This is *not* a causal estimate — there is no control group, and notes are written at dips, which revert (regression to the mean). What it does establish is worse: **from current logs, contact effectiveness is unknowable** — which is exactly the argument for the one-field outcome log ("did it land?") in the roadmap.

- **Note-reader, measured against humans:** all 75 note threads were hand-labelled from the notes alone, against a written codebook, *before* the extractor ran (`eval/`). v3.1 agrees with the gold labels at $\kappa = 0.80$ (87% strict, 97% lenient) — and the eval earned its keep by catching a real bug first: v3 read "vanished + unreachable parents" as `needs_help` (failing-recall 0.15); one prompt revision fixed it.
- **Closing the loop:** 6 surfaced students were *already contacted and still declined* — they do not need another drafted message; they are flagged for **escalation beyond messaging** (S005, S023, ...).

- **Measurement without a holdout:** denying triage to a live classroom is ethically wrong and, at one cluster, statistically empty. Instead, three pre-registered designs (`docs/EVAL_PLAN.md`): a Day-20 primary readout (failer contact rate vs the measured 44% baseline), a regression discontinuity at the capacity cutoff (the top-k call list creates near-identical students on either side of an arbitrary line), and a stepped-wedge rollout for 20 → 100 campuses — the deployment order *is* the experiment.

7.1 Fairness

Guarding against the documented early-warning harms

The risk score uses *behaviour only* (attendance, practice, quiz, trajectory) — never `learning_track`, `campus`, name, phone, or grade. The audit reports surface-rate by track (Remedial is surfaced more because it is genuinely failing more, by behaviour, not by membership), and a *within-track* report view keeps the most-at-risk Standard/Accelerated students visible for review — and Remedial membership never becomes a self-entrenching re-flag. The audit also tracks the equity axis specific to a notes-reading system: per-facilitator note coverage (21–55%, and the largest classroom is the least documented), so note-informed care cannot silently concentrate where documentation happens to be best. This directly answers the real-world failures where recall-maxed dropout models concentrated false alarms on the most disadvantaged with no benefit.

8 The evidence base

The design is not invented; each choice traces to the early-warning and behavioural-nudge literature.

- **Prediction without action fails.** A 73-school RCT of an early-warning system improved intermediate proxies but *not* the ultimate outcome; a 38-study review found dashboards do not improve achievement. *So we built an action layer, and measure effectiveness.*
- **Personalised, parent-directed nudges causally work**, concentrated on the dropout / re-engagement margin and among the highest-risk students; improvement-focused content beats praise. *So the message is parent-directed, improvement-focused, one step.*
- **The trusted local sender is load-bearing** — nudges that worked locally collapsed to a precise null at national scale when the sender became impersonal. *So the facilitator sends from their own name; we never centralise to "Boon".*
- **Arabic LLMs read dialect well but generate it poorly.** *So we extract with the model and draft in warm MSA.*

Confirmed effect sizes are modest (single-digit percentage points). **None** of the causal evidence is Saudi or Arabic — external validity for Boon's context is a hypothesis to pilot, which is exactly why the rollout is pre-registered as a stepped wedge rather than assumed to transfer. "Two-way beats one-way" and "nudges survive at scale" did not survive verification and are treated as open.

9 Results and how quality is measured

- **Runs end-to-end**, one command, LLM on; committed `outputs/` (queue + Arabic drafts).
- **22/23 acceptance tests** pass — the demo cast plus the system invariants: every quiz-failer leaves with a ready action; the cap binds only calls/1-on-1s; the faithfulness gate blocks unevidenced and fabricated claims; the Day-9 backtest stays predictive.
- **Ranking backtested:** scored on days 1–9 only (pre-quiz clock), the risk score ranks who actually failed Quiz 1 at **AUC 0.911** (precision@66 = 0.85).
- **LLM lift:** the notes changed the outcome for **18 of 75** noted students vs. rules-only (9 escalated, 8 de-escalated to hand capacity back, 1 to human review).
- **Note-reader vs. human gold labels:** $\kappa = 0.80$, 87% strict / 97% lenient ($n = 75$, single-rater, codebook-adjudicated).
- **Contact outcomes (descriptive):** 11% re-engaged / 11% declined / 13% logged too late, of 75 — reported with its limitations (no control group; regression to the mean).
- **Faithfulness** (evidence must be a real substring) and **abstention** (low confidence → human) guard the AI; quality is reproducible (temp 0 + content-hash cache).

10 Limitations and what I would build next

The decision logic is tuned against a 200-student sample (it is both the design target and the test set); the gold labels are single-rater (a second Arabic-speaking rater is the planned upgrade to true inter-rater κ); the effectiveness "no-change" bucket includes stable passing students, so those percentages are directional; and the causal designs (RD at the cutoff, stepped wedge) are pre-registered but unrun until Quiz-2 data exists.

Next, gated on richer data and scale: a per-student *temporal narrative* (the shape of the decline, not a tier); *knowledge tracing* to diagnose the specific concept a student fails (needs per-question data); a *student-voice* two-way check-in; and a causal *what-works-for-whom* loop (uplift / bandits) once volume justifies it. The single most important addition is the one-field outcome log ("did the contact land?") plus per-question quiz instrumentation — moving the metric from *attempted* to *effective*, and the diagnosis from "at risk" to "weak on quadratics".